

Choosing a Ratio Representation

Extending an equivalent-ratio table, finding a unit rate, scaling a ratio up with a proportion, and comparing two rates to decide which is better

Directions.

- You have four tools: a ratio **table**, a **unit rate**, a **proportion**, and a direct **comparison**. Pick one, and write down which one you picked before you compute.
- Equivalent ratios come from *multiplying* both parts by the same number. Adding the same number to both parts changes the ratio.
- When two rates are being compared, put them in the same form first — usually “per one of something”.
- Round money answers to the cent.

1. A trail-mix recipe mixes oats and raisins in the ratio 3 cups of oats to 2 cups of raisins. Complete the ratio table.

Oats (cups)	3	6	9
Raisins (cups)	2		

Then use the table to say how many cups of raisins go with 12 cups of oats, and write down what you multiplied by to get there.

Answer: _____

- 2.** A five-pack of notebooks costs 12.50 dollars. Find the cost of one notebook, in dollars. Say why a unit rate is the useful representation here.

Answer: _____

- 3.** A printer prints 14 pages every 4 minutes. Complete the ratio table.

Minutes	4	8	12	20
Pages	14			

Which column did you find easiest, and why?

Answer: _____

4. On a hiking map, 4 centimetres represents 5 kilometres. Write and solve a proportion to find how many centimetres represent 15 kilometres.

$$\frac{x}{15} = \frac{4}{5}$$

Answer: _____

5. The corner store sells notebooks 5 for 12.50 dollars. The supermarket sells the same notebooks 6 for 17.40 dollars.
Which is the better buy? Show the two rates you compared, and write the comparison as a statement using $<$ or $>$.

Answer: _____

6. A car travels 174 miles on 6 gallons of fuel. Find the fuel economy in miles per gallon.

Answer: _____

7. Sam is extending the ratio 2 cups of concentrate to 5 cups of water. For 4 cups of concentrate he wrote 7 cups of water, “because I added 2 to each side”.

(a) Explain what is wrong with adding to both parts.

(b) Give the correct number of cups of water for 4 cups of concentrate.

Answer: _____

8. A soup recipe for 8 servings uses 20 ounces of tomatoes. Write and solve a proportion for the ounces of tomatoes needed for 14 servings.

$$\frac{x}{14} = \frac{20}{8}$$

Answer: _____

9. Runner A covers 9 kilometres in 40 minutes. Runner B covers 7 kilometres in 32 minutes. Who is running faster? Choose a representation, show both rates in that same form, and write the comparison using $<$ or $>$.

Answer: _____

10. Three gallons of paint cover 1050 square feet. A wall needs 1750 square feet of coverage. Find the unit rate first, then use it to find how many gallons are needed.

Answer: _____

11. Team A has won 6 of its 9 games. Team B has won 10 of its 15 games. Which team has the better winning ratio? Be careful — write both ratios in the same form before you decide, and state the comparison using $<$, $>$ or $=$.

Answer: _____

12. Challenge. Two lemonade recipes: Recipe R mixes 2 cups of concentrate with 5 cups of water. Recipe S mixes 3 cups of concentrate with 8 cups of water.

(a) Which recipe tastes stronger? Compare concentrate per cup of water, and state the comparison using $<$, $>$ or $=$.

Answer: _____

(b) How many cups of concentrate does Recipe R need for 35 cups of water? _____

(c) How many cups of concentrate does Recipe S need for 24 cups of water? _____